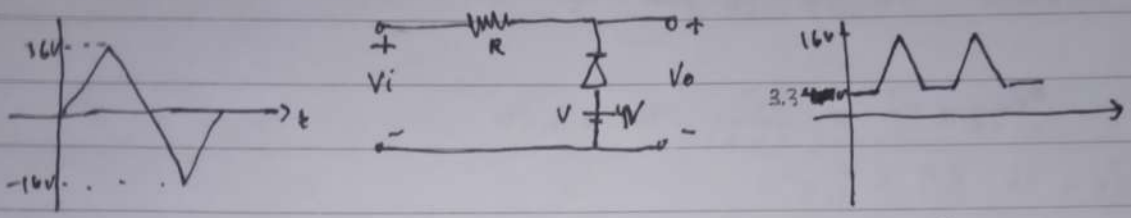
Nama : Andrean Noriansyah
NIM : 24134400 4
KELAS : 2A - TNK

No
Date

No. 1

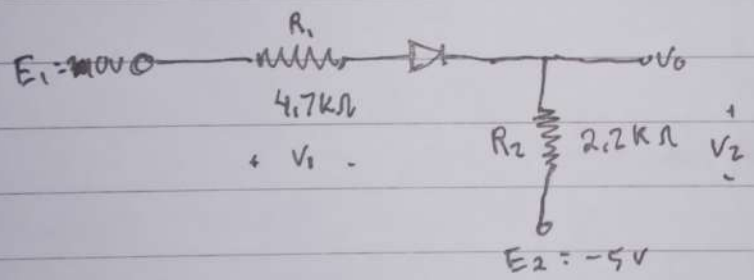
Gambarkan bentuk golombang output dari rangkaian clipper paralel pada rangkaian berikut. Apabila dioda pada rangkaian tersebut menggunakan dioda silikon, perhitungkan pula tegangan bias majunya (tegangan bias maju dioda silikon sebesar 0,7 V)



Tegangan bias maju dioda silikon = 4V - 0,7V = 3,3V

No. 2

Tentukan besar I, V_1, V_2, dan V_o pada rangkaian seri dc dari dioda silikon berikut.



$-E_1 + IR_1 + V_D + IR_2 - E_2 = 0$

$-10 + I4,7k + 0 + I2,2k - 5 = 0$

$-15 + I6,9K = 0$

$I = \frac{15}{6,9k} = 2,73\ mA$

$V_1 = I \cdot R_1 = 2,73\ mA \cdot 4,7k = 10,217\ V$

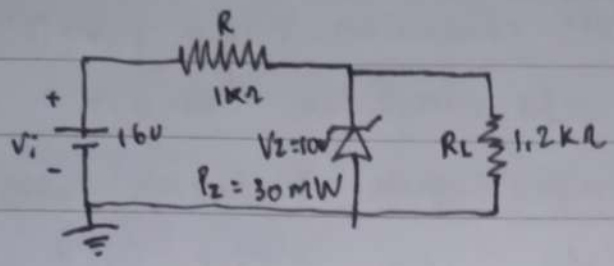
$V_2 = I \cdot R_2 = 2,73\ mA \cdot 2,2k = 4,783\ V$

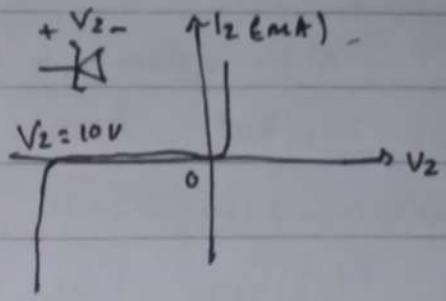
$V_o = 10 - 0,7 = 9,3\ V$

No

Date

No. 3





1. Tentukan V_L, V_R, I_Z dan P_Z

$$V_L = \frac{R_L}{R_L + R} \times V_i = \frac{1{,}2}{1{,}2 + 1} \cdot 16 = 8{,}7\,V$$

$$V_R = 16 - 8{,}7 = 7{,}3\,V$$

$I_Z = 0$ karena $V_R \leq V_Z$

2. Ulangi no. 1 jika R_L diganti dengan 3kΩ

$$V_L = \frac{R_L}{R_L + R} \times V_i = \frac{3}{3+1} \cdot 16 = 12\,V$$

karena $V_L > V_Z \rightarrow V_L = V_Z = 10V$

$V_R = 16 - 10 = 6V$

$I = 6mA$

$I_L = \frac{10}{3k} = 3{,}33\,mA$

$I_Z = 2{,}67mA$